Lab 8: HTTP, JSON Parsing & Permissions · Debugging Report (BuggyReader)



After both fixes

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Repo: `https://github.com/NeitLN/Lab8_NetworkJSON_2474802010391` (folder `BuggyReader/`)

Device: Pixel 7 emulator, Android 14 (API 34) · Logcat level Error, tag `AndroidRuntime`

Commits: `ef641c7` buggy version as committed, `b48975a` fix #1, `56b9b2b` fix #2

Steps I followed: run the app and tap Load posts (crash), copy the trace from `FATAL EXCEPTION` to the first `com.vlu.buggyreader` line, fix defect #1, run again (a different crash), fix defect #2, run again. After fix #2 the TextView shows all 100 titles and nothing is printed in the crash buffer (screenshot on the right).

#	Exception class	Key Logcat line	Root cause (own words)	Fix applied (file and line)	AI used? (prompt and what you changed)
1	`android.os.NetworkOnMainThreadException`	`at com.vlu.buggyreader.MainActivity.loadPosts(MainActivity.kt:25)` (just below `StrictMode$AndroidBlockGuardPolicy.onNetwork`)	`loadPosts()` is called straight from the click listener, so `connection.inputStream` opens the socket on the Main (UI) thread. Since API 11 StrictMode stops any network call on that thread, because a slow request would freeze the screen and cause an ANR after about 5 s.	`MainActivity.kt` L28 to 40: `lifecycleScope.launch` (L30) + `withContext(Dispatchers.IO) { downloadTitles() }` (L32); the blocking code moved to `downloadTitles()` (L43 to 59); `tvResult` is set back on Main (L33). `app/build.gradle.kts` L42 to 43: lifecycle-runtime-ktx, coroutines-android.	
2	`java.lang.SecurityException`	`SecurityException: Permission denied (missing INTERNET permission?)` `at ...MainActivity.downloadTitles(MainActivity.kt:48)`	The manifest has no `android.permission.INTERNET`, so the app is not allowed to open sockets and the DNS lookup of jsonplaceholder.typicode.com is denied. It only showed up after fix #1, because before that StrictMode stopped the call before the lookup ran. INTERNET is a normal permission: declaring it is enough, no runtime dialog.	`AndroidManifest.xml` L6: `<uses-permission android:name="android.permission.INTERNET" />` as a direct child of `<manifest>`, outside `<application>`.	

Stack trace #1 (before any fix, tap Load posts)

```
FATAL EXCEPTION: main
Process: com.vlu.buggyreader, PID: 25636
android.os.NetworkOnMainThreadException
    at android.os.StrictMode$AndroidBlockGuardPolicy.onNetwork(StrictMode.java:1675)
    at java.net.Inet6AddressImpl.lookupHostByName(Inet6AddressImpl.java:115)
    at java.net.Inet6AddressImpl.lookupAllHostAddr(Inet6AddressImpl.java:103)
    at java.net.InetAddress.getAllByName(InetAddress.java:1152)
    at com.android.okhttp.Dns$1.lookup(Dns.java:41)
    at com.android.okhttp.internal.http.RouteSelector.resetNextInetSocketAddress(RouteSelector.java:178)
    at com.android.okhttp.internal.http.RouteSelector.nextProxy(RouteSelector.java:144)
    at com.android.okhttp.internal.http.RouteSelector.next(RouteSelector.java:86)
    at com.android.okhttp.internal.http.StreamAllocation.findConnection(StreamAllocation.java:176)
    at com.android.okhttp.internal.http.StreamAllocation.findHealthyConnection(StreamAllocation.java:128)
    at com.android.okhttp.internal.http.StreamAllocation.newStream(StreamAllocation.java:97)
    at com.android.okhttp.internal.http.HttpEngine.connect(HttpEngine.java:289)
    at com.android.okhttp.internal.http.HttpEngine.sendRequest(HttpEngine.java:232)
    at com.android.okhttp.internal.huc.HttpURLConnectionImpl.execute(HttpURLConnectionImpl.java:465)
    at com.android.okhttp.internal.huc.HttpURLConnectionImpl.getResponse(HttpURLConnectionImpl.java:411)
    at com.android.okhttp.internal.huc.HttpURLConnectionImpl.getInputStream(HttpURLConnectionImpl.java:248)
    at com.android.okhttp.internal.huc.DelegatingHttpsURLConnection.getInputStream(DelegatingHttpsURLConnection.java:211)
    at com.android.okhttp.internal.huc.HttpsURLConnectionImpl.getInputStream(HttpsURLConnectionImpl.java:30)
    at com.vlu.buggyreader.MainActivity.loadPosts(MainActivity.kt:25)
```

Stack trace #2 (after fix #1, tap Load posts again)

```
FATAL EXCEPTION: main
Process: com.vlu.buggyreader, PID: 26251
java.lang.SecurityException: Permission denied (missing INTERNET permission?)
    at java.net.Inet6AddressImpl.lookupHostByName(Inet6AddressImpl.java:150)
    at java.net.Inet6AddressImpl.lookupAllHostAddr(Inet6AddressImpl.java:103)
    at java.net.InetAddress.getAllByName(InetAddress.java:1152)
    at com.android.okhttp.Dns$1.lookup(Dns.java:41)
    at com.android.okhttp.internal.http.RouteSelector.resetNextInetSocketAddress(RouteSelector.java:178)
    at com.android.okhttp.internal.http.RouteSelector.nextProxy(RouteSelector.java:144)
    at com.android.okhttp.internal.http.RouteSelector.next(RouteSelector.java:86)
    at com.android.okhttp.internal.http.StreamAllocation.findConnection(StreamAllocation.java:176)
    at com.android.okhttp.internal.http.StreamAllocation.findHealthyConnection(StreamAllocation.java:128)
    at com.android.okhttp.internal.http.StreamAllocation.newStream(StreamAllocation.java:97)
    at com.android.okhttp.internal.http.HttpEngine.connect(HttpEngine.java:289)
    at com.android.okhttp.internal.http.HttpEngine.sendRequest(HttpEngine.java:232)
    at com.android.okhttp.internal.huc.HttpURLConnectionImpl.execute(HttpURLConnectionImpl.java:465)
    at com.android.okhttp.internal.huc.HttpURLConnectionImpl.getResponse(HttpURLConnectionImpl.java:411)
    at com.android.okhttp.internal.huc.HttpURLConnectionImpl.getInputStream(HttpURLConnectionImpl.java:248)
    at com.android.okhttp.internal.huc.DelegatingHttpsURLConnection.getInputStream(DelegatingHttpsURLConnection.java:211)
    at com.android.okhttp.internal.huc.HttpsURLConnectionImpl.getInputStream(HttpsURLConnectionImpl.java:30)
    at com.vlu.buggyreader.MainActivity.downloadTitles(MainActivity.kt:48)
```

Note on trace #2: the frames under it are `kotlinx.coroutines.scheduling.CoroutineScheduler$Worker`, so the lookup really ran on an IO worker thread (fix #1 worked). The header still says `main` because the exception is not caught inside the coroutine, so it goes back up to `lifecycleScope` on the Main thread and crashes the app there. Full traces: `BuggyReader/docs/logcat/`.